Structured Localization and Auxiliary Adaptation for Japanese Grammatical Error Correction

Abstract

Local grammatical correction requires both a source location and a replacement. We study a category-conditioned Japanese correction system that fixes the interval before revision, combines recurrent and sparse local predictors, and selects between a compact editor and an adapted pretrained generator through a likelihood-based fallback. The main experiment compares source-only and auxiliary-plus-source adaptation of the same four-billion-parameter instruction model, with matched target fitting, frontend and decoding. On a reused benchmark with 200 development and 559 historical test examples, three-seed mean exact span-and-replacement agreement (ExactMatch) rises from 27.97% to 33.04%. The gain is 5.07 percentage points, with a six-metric-adjusted paired 95% interval of 1.18–9.32 points; localization ties and incremental partial-similarity intervals include zero. The original character gated recurrent unit and fast fusion reach 14.31% and 17.05%, respectively. The study retains retrieval and supervised baselines, unsuccessful controls, a surface-probability decoding diagnostic and measured local-compute costs. An 803-case external localization projection reveals severe transfer failure, while earlier model families obtain zero exact corrections on a reserved 25-case style complement. The four-billion-parameter model has not been tested on that complement. The results support a bounded retrospective improvement, while source overlap, adaptive benchmark reuse and external failures leave independent correction generalization unresolved.

Keywords: Japanese grammatical error correction; structured prediction; auxiliary training; constrained decoding; low-rank adaptation

1 Introduction

Local grammatical correction requires two decisions: identifying the intended error and producing its replacement. A fluent sentence rewrite can fail the first decision, while a correct location can still receive an unsuitable revision. This distinction matters for annotation-aligned correction, where the system must change one designated fragment and preserve the surrounding text.

We study Japanese correction with a supplied error category and limited source supervision. The benchmark has 200 development examples and 559 historical test examples. A 97,833-parameter character gated recurrent unit (GRU) provides a compact starting point. Sparse localization and editing models improve this system, and an adapted pretrained generator supplies an alternative replacement for the same predicted interval. The method separates localization from generation and applies a fixed likelihood comparison to choose the final replacement.

The principal experiment compares source-only and auxiliary-plus-source adaptation of the same Qwen3-4B-Instruct-2507 checkpoint. Both arms use the same context frontend, target schedule and decoder. Auxiliary exposure raises three-seed mean ExactMatch from 27.97% to 33.04% on the historical benchmark. The paired six-metric-adjusted interval for this gain is positive; two partial-similarity intervals include zero. This matched comparison is more informative about auxiliary adaptation than the comparison with the original GRU, which also changes capacity, frontend and pretraining.

The contribution is an implemented correction system and a controlled empirical study of its components, auxiliary training and resource costs. An openly licensed metadata-and-numerical-results dataset accompanies the release package, with 759 metadata records and 10,062 per-example evaluation records from six methods and three seeds. A separate surface-probability recurrence provides a checkable alternative to edit-path scoring. GRUs, probability mixtures, edit representations, low-rank adaptation and probability-mass bounds are established techniques; their combination here is not claimed as a new general learning algorithm. The historical benchmark has been reused during development and contains source overlap. External localization and a small reserved style evaluation expose serious transfer failures. These findings bound the contribution to retrospective annotation agreement rather than general Japanese correction or state of the art.

2 Related work

Grammatical error correction includes sentence generation, edit tagging and different evaluation conventions [1, 2]. Japanese research includes correction corpora, human quality-estimation resources and error-aware example retrieval [3–5]. A local edit, supplied category or retrieved correction is therefore not itself a novel contribution.

Homma and Komachi examine non-autoregressive Japanese correction and CPU latency with a Levenshtein Transformer [6]. Position-offset and copy-based Japanese correction have also been studied [7]. Their datasets and sentence-level metrics differ from the present one-span task, so their published scores cannot be numerically ranked against our ExactMatch. Likewise, mapping a released sentence corrector to one category-conditioned edit changes its task.

The present components draw on recurrent encoders [8], lexical retrieval [9], low-rank adaptation [10] and input-dependent mixtures [11]. Weighted finite-state methods and connectionist temporal classification establish the distinction between path and output probability [12, 13]. Neural finite-state combination and edit-level voting also precede this work [14, 15]. Our claims concern the specified implementation and controlled results, rather than invention of these principles.

3 Task, data and evaluation

3.1 Correction contract and source data

For a character sequence x and supplied category c, the system predicts one nonempty interval $[s, e)$ and replacement y. The interval has at most 40 characters. A productive replacement is nonempty, differs from $x_{s:e}$, and has length at most $\max(44, 4(e-s))$. The surrounding text is preserved. Category prediction, discovery of all errors and full-document proofreading are outside the task.

CAMC-JL contains 759 annotation-derived examples from local Natane and ICJS/TUFS snapshots: 200 development and 559 historical test cases. Its ten categories are aspect, tense, style, passive, transitivity, honorific, benefactive, demonstrative, sentence connection and causative. Retained windows contain at least 55 characters. ICJS/TUFS context extends up to 90 characters on each side before filtering. Each reference is linked to a source annotation, and the original dataset is preserved.

Development examples form 121 connected components under available source-document, learner and window identities. Three fixed folds contain 67, 67 and 66 held-out examples. Fitting inventories and vocabularies are rebuilt within each fitting fold. These folds have been reused across studies and are not fresh validation. The historical test was inspected in earlier work: 313 cases share a development document and 60 contain a typed correction pair seen in development. Available identities define 194 test clusters and a 207-case subset without a shared available source unit. The original split does not establish complete learner separation.

3.2 Auxiliary and diagnostic resources

A public synthetic resource supplies 6,297 valid teacher-authored simulated error pairs [16]. Connected sentence families divide these into 5,093 training, 636 validation and 568 reserved examples. A separately sampled automatic resource contributes 54,674 eligible pairs from 43,423 families. Declared normalized-text overlap rules exclude protected original and reserved texts, but do not establish semantic or pretrained-base decontamination.

The matched 4B auxiliary arm uses 16,000 distinct pairs: 1,389 teacher-authored and 14,611 automatic pairs from the 59,767-pair eligible training pool. These are simulated errors, not 16,000 new learner observations. The 568 reserved teacher examples do not enter the reported evaluation. Their supplied-span task contract differs from the full category-conditioned localization task.

Two other resources assess limits rather than enlarge the primary correction test. A Teramura projection contains 803 localization-only records from 221 learners across nine categories [17]; its abstract correction labels do not supply literal replacement references. A reserved Natane style complement contains 25 literal corrections from 14 learners absent from the earlier Natane learner set. These labels were inspected for construction but excluded from fitting and model selection. A further 61 supplied-span pairs remain unused. The supplementary technical record documents construction, exclusions and provenance.

3.3 Metrics and inference

Six example-mean metrics distinguish localization from revision. SpanMatch (SM) is exact source-interval agreement; SpanOverlap (SO) is interval intersection over union. RevisionMatch (RM) is exact replacement agreement

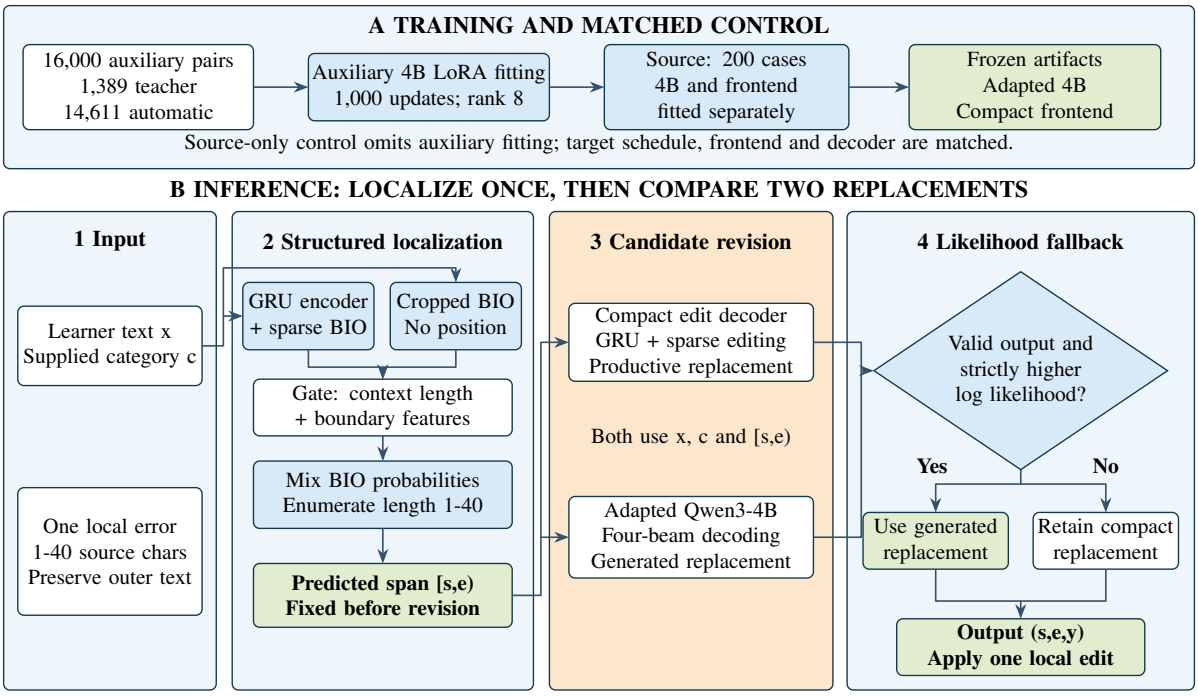


Fig. 1 Training and inference pipeline. The source-only control omits auxiliary LoRA fitting and retains the same target-fitting schedule, frontend and output rule. Source fitting trains the 4B adapter and compact frontend separately. The context gate combines original and crop-trained BIO predictors; a single interval is fixed before either editor is queried. The generated replacement is accepted only when it satisfies the output constraints and strictly exceeds the compact replacement in length-normalized log likelihood, including the end token. Otherwise the compact replacement is retained. No gold span is supplied at inference.

after the specified Unicode/whitespace normalization, and RevisionSimilarity (RS) is one minus normalized character Levenshtein distance. SoftMatch is SO times RS; ExactMatch is SM times RM. RM and ExactMatch coincide for some systems but have different definitions.

The legacy scorer resolves a predicted substring appearing several times by maximizing gold-span overlap, then minimizing distance from the gold start. Its localization scores can therefore be optimistic for repeated substrings. Saved-offset diagnostics accompany these scores. The external localization evaluation uses actual predicted offsets directly. Agreement with one reference is not a human rating of fluency or correction quality.

Within a declared study, selection maximizes the minimum of six score ratios to its incumbent, then the mean ratio, ExactMatch, SoftMatch and frozen tie rules. Later studies require non-regression on original and pooled stress means; stress crops remain transformations of the same development parents. The completed sequence contains 272 valid configurations in 21 development studies. All declared candidates and unsuccessful controls remain in the supplementary record. This extensive adaptive reuse contributes selection uncertainty.

Final target fits use seeds 20260927, 20260928 and 20260929. For paired historical comparisons, seed-matched differences are averaged within each example before 20,000 bootstrap resamples of 194 available-source-unit clusters. Simultaneous 95% intervals adjust six endpoints within each comparator/subset family, not all adaptive studies. These intervals condition on fitted models and retained examples; source clusters do not establish full learner independence.

4 Correction system

Figure 1 summarizes training and inference. The matched comparison changes auxiliary exposure for the 4B adapter. The compact frontend is fitted separately on source supervision; at inference, one predicted interval is shared by both replacement candidates.

4.1 Recurrent and sparse fusion

The original bidirectional GRU uses 48-dimensional character embeddings, 16-dimensional category embeddings and 48 hidden units per direction. BIO, character-edit and prefix heads share its representation. Training combines whole-span negative log-likelihood with weighted BIO and edit losses, using AdamW, batch size 16, dropout 0.1, gradient clipping at 1, five warm-up epochs and cosine decay over 80 epochs.

The sparse localizer uses characters within radius three, local bigrams/trigrams, category interactions, script identity and coarse relative position. The editor also uses interval length, within-span position and boundary features. Features are L2-normalized. Logistic-loss stochastic gradient descent uses L2 regularization 10^{-4}, at most 2,000 iterations and tolerance 10^{-4}; BIO fitting balances classes.

After aligning O/B/I classes, average the recurrent and linear distributions: $q_i(k) = [p_G(k \mid x,c,i) + p_L(k \mid x,c,i)]/2$. The score of an interval relative to the all-O path is

$$S(s,e) = \log \frac{q_s(B)}{q_s(O)} + \sum_{i=s+1}^{e-1} \log \frac{q_i(I)}{q_i(O)}. \tag{1}$$

Enumeration over lengths 1–40 costs $O(40|x|)$; ties prefer shorter, then earlier intervals. Localization is completed before revision search.

Minimum-edit alignment assigns a prefix emission and one action per source character. Actions keep, delete or emit a learned string. The editor averages aligned component distributions over the union of their fitting-only label inventories. Best-first search examines at most 256 states and returns the highest-ranked encountered productive replacement. If none is found, it emits an empty prediction instead of moving to another source interval. The fast fusion stores 97,833 neural parameters and 3,467,721 linear coefficients/intercepts, totaling 3,565,554 values.

4.2 Context-dependent localization

An additional compact frontend blends the original localizer with one fitted on cropped contexts without coarse position features. Within each fitting fold, balanced L2 logistic regression predicts whether the context was shortened, using log length, its square, whitespace fraction, final punctuation and leading/trailing whitespace flags. The resulting continuous weight mixes original and crop-trained BIO probabilities. Gate inputs contain no reference position, category label, identity or model prediction.

Training crops use widths 32/64 and left-context fractions $0, 1/3, 2/3, 1$; held-out stress crops use $1/4, 3/4$. Gold spans are used to construct development crops, but inference receives only text and category. The editor remains fixed. Selection uses original and pooled stress means, not every individual view. The resulting frontend stores 4,234,774 values. It improves the constructed stress condition but does not establish reliable transfer to naturally short text.

4.3 Matched adaptation and fixed-span fallback

The principal comparison uses pinned Qwen3-4B-Instruct-2507 weights [18], with 4,022,468,096 base parameters and 16,515,072 trainable rank-8 low-rank adaptation parameters. Including the context frontend gives 4,043,217,942 stored model values. This is a local-GPU system, distinct from the small CPU fusion.

The source-only arm starts from the untrained adapter. The auxiliary arm first receives 1,000 effective-batch-16 updates on the fixed 16,000 auxiliary pairs, then starts a fresh target optimizer. Both arms share the source folds, frontend, ten-epoch target fit, thirty-epoch learning-rate horizon and decoder. Target fitting uses AdamW with peak learning rate 10^{-4}, 5% warm-up, cosine decay, weight decay 0.01 and clipping at 1. Microbatches of two accumulate to an effective batch of 16. Only response tokens contribute to the supervised loss. Each full-development refit makes 130 updates across ten epochs under a 390-update horizon.

The prompt requests only a replacement for the predicted marked interval. Four-beam deterministic decoding generates a candidate; the gold interval never reaches inference. For replacement tokens $t_1, \ldots, t_m$, including the end token, define

$$\ell(y \mid x,c,s,e) = m^{-0.6} \sum_{j=1}^{m} \log p(t_j \mid t_{<j}, x, c, s, e). \tag{2}$$

The system adopts the generated replacement only if it ends correctly, satisfies the productive constraints without internal line breaks, and has a strictly greater score than the compact editor's replacement. Otherwise it retains the compact replacement and its interval. Leading/trailing whitespace is stripped; no gold-dependent output repair is applied.

The principal runs use a BF16 base, float32 adapters and generation/scoring batches of four. Prior controls document sensitivity to batch composition and arithmetic precision. Finite seed/annotation-field canaries and exact archived replay checks do not imply bitwise equivalence across hardware or arbitrary request batches.

4.4 Surface-probability diagnostic

A separate decoder study distinguishes the most likely edit path from the most likely resulting string. Merge equivalent slot emissions into $q_i(a)$. For a candidate y, let $F_i(j)$ be the mass with which the first i slots emit its first j characters, with $F_0(0) = 1$. For an emission matching y at offset j,

$$F_{i+1}(j + |a|) \mathrel{+}= F_i(j) q_i(a). \tag{3}$$

Empty emissions advance the slot only. The final value sums all paths yielding y, including unenumerated paths.

A length recurrence gives productive output mass V. For distinct scored candidates C, residual mass is $R = V - \sum_{y \in C} P(y)$. Every unseen productive string has probability at most R, so the best known candidate is globally productive maximum a posteriori when its probability exceeds R, assuming exact arithmetic. Float64 checks with a 10^{-12} margin provide a numerical certificate, not an interval-arithmetic guarantee or a linguistic-quality guarantee. This diagnostic is separate from the selected 4B output rule.

5 Results

5.1 Historical baseline comparison

Table 1 retains retrieval, supervised and adapted baselines on the same 559 historical examples. Frequency memory, TF–IDF and BM25 retrieve fitting-derived corrections; the linear systems separate sparse localization from nearest-revision retrieval or edit tagging. Neural rows average three fixed seeds.

Table 1: Historical annotation agreement (%). Neural systems average three seeds; retrieval and linear systems are deterministic. The test is reused and source-overlapping.

Method	SM	SO	RM	RS	Soft	Exact
Frequency memory	10.73	12.59	8.77	15.88	9.54	8.23
TF–IDF retrieval	12.88	14.39	10.38	16.98	11.54	10.02
BM25 retrieval	13.06	14.52	10.55	16.92	11.57	10.20
Linear + nearest revision	66.91	74.28	12.16	33.27	28.86	8.41
Linear edit tagger	66.91	74.28	9.84	44.13	41.79	9.84
Character GRU	66.79	67.85	14.31	40.84	38.10	14.31
Fast GRU–linear fusion	75.97	77.50	17.05	46.71	45.29	17.05
Auxiliary-16k, 0.6B	75.97	77.50	21.65	44.72	43.28	21.65
Coverage-37,888, 0.6B	75.97	77.50	22.12	45.80	44.34	22.12
Auxiliary-16k, 1.7B	75.97	77.50	26.77	48.89	47.53	26.77
Context gate, 1.7B	76.45	77.90	26.71	49.00	47.71	26.71
Context gate, source-only 4B	76.45	77.90	28.15	51.06	49.03	27.97
Context gate, auxiliary 4B	76.45	77.90	33.09	53.26	51.77	33.04

Fusion raises ExactMatch from 14.31% to 17.05% and improves all six metrics over the GRU with positive simultaneous paired intervals. Its ExactMatch difference is 2.74 percentage points [1.26,4.44]. Localization fusion alone reaches 15.80% and revision fusion alone 15.44%. Productive decoding raises ExactMatch from the fused argmax decoder's 16.16% to 17.05%, but lowers SoftMatch from 46.54% to 45.29%. Development selection does not imply historical dominance on every metric.

Table 2 separates the compact components. Localization fusion adds 9.18 points of SpanMatch over the GRU, but only 1.49 points of ExactMatch; revision fusion adds 1.13 points of ExactMatch at unchanged localization. Better localization therefore exposes a substantial replacement problem. Productive search further trades partial agreement for exact agreement, consistent with its task-specific output constraint rather than uniform improvement in string similarity.

Figure 2 shows why a stronger replacement model remains useful after fusion. For the auxiliary 4B pipeline, 43.41% of examples have an exact location but an incorrect replacement, while 23.55% have an incorrect location. The matched source-only model has the same localization error share and a larger 48.48% correctly localized but incorrectly revised share. ExactMatch divided by SM rises from 36.58% to 43.21%; these ratios of aggregate

Table 2: Frozen component ablations on the historical test. Productive decoding improves exact correction over argmax but lowers partial similarity.

Method	SM	SO	RM	RS	Soft	Exact
GRU	66.79	67.85	14.31	40.84	38.10	14.31
Linear, productive	66.91	74.28	11.63	43.21	41.19	11.63
Localization fusion	75.97	77.50	15.80	44.52	43.11	15.80
Revision fusion	66.79	67.85	15.44	42.90	40.10	15.44
Fusion, argmax	75.97	77.50	16.16	48.16	46.54	16.16
Fusion, productive	75.97	77.50	17.05	46.71	45.29	17.05

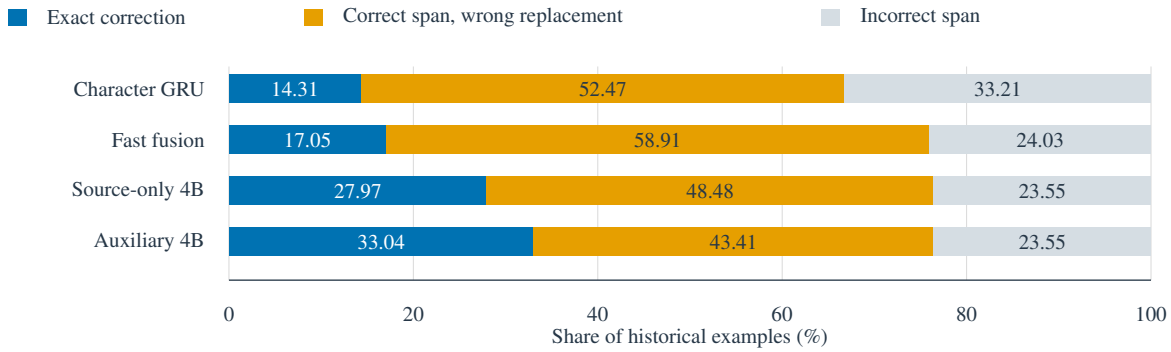


Fig. 2 Historical error decomposition from three-seed aggregate means. Exact correction is ExactMatch; correct location with an incorrect replacement is SM minus ExactMatch; an incorrect location is one minus SM. The three disjoint scoring outcomes sum to 100%. This is an algebraic decomposition under the retained legacy scorer, not a new human error annotation or an independent evaluation.

means describe replacement agreement among correctly localized outputs. They are not scores from a gold-span experiment. The larger intermediate error share for fast fusion than for the GRU reflects successful localization of more cases without a corresponding exact replacement, rather than a decline in overall exact correction.

Smaller adapted systems expose a similar tradeoff: auxiliary 0.6B correction reaches 21.65% ExactMatch while its partial-similarity scores remain below fast fusion. The auxiliary 1.7B model reaches 26.77%. Capacity, instruction tuning, frontend and training schedules differ across some rows, so the table is not an isolated scaling experiment.

5.2 Matched 4B auxiliary comparison

On reused development data, auxiliary-plus-source fallback raises original ExactMatch from 25.50% to 27.50% and pooled stress ExactMatch from 21.875% to 24.875%, with tied localization. A 0.0766-point RS decline in one left-positioned 64-character crop remains; the candidate is not uniformly better in every view. Development selection fixes the primary before its subsequent historical prediction.

On the historical benchmark, auxiliary adaptation raises ExactMatch by 5.069 percentage points relative to the matched source-only 4B control (Table 3; Fig. 3). RM and ExactMatch intervals are positive, whereas RS and SoftMatch intervals include zero. Localization ties by construction. Against the GRU, all six intervals are positive; ExactMatch increases by 18.724 points [14.045,23.722]. All four revision intervals are positive against fast fusion and both adapted 1.7B comparators.

Auxiliary ExactMatch ranges from 32.200% to 33.631% across the three seeds, versus 27.549%–28.265% for source-only adaptation. These are descriptive ranges, not intervals over future training runs. Each seed has four revision-mean gains and tied localization on the full set. However, the 60 correction-pair-seen cases regress: RM/ExactMatch fall by 1.6667 points, RS by 2.3796 and SoftMatch by 2.1019. Full-set improvement does not imply subgroup dominance.

All 1,677 new and 8,385 archived comparator outputs pass both legacy scorers and saved-offset checks. The audit reconstructs 1,677 new and 5,031 archived neural fallback decisions, and all 36 paired seed/annotation-field canaries pass. These checks support implementation consistency without removing evaluation dependence.

Table 3: Auxiliary minus source-only 4B on the historical test. Intervals adjust six metrics within this comparison; they do not adjust the full adaptive study sequence.

Metric	Difference (percentage points)	Simultaneous 95% interval
SpanMatch	0.000	[0.000, 0.000]
SpanOverlap	0.000	[0.000, 0.000]
RevisionMatch	4.949	[1.068, 9.240]
RevisionSimilarity	2.193	[-0.795, 5.397]
SoftMatch	2.738	[-0.054, 5.823]
ExactMatch	5.069	[1.179, 9.324]

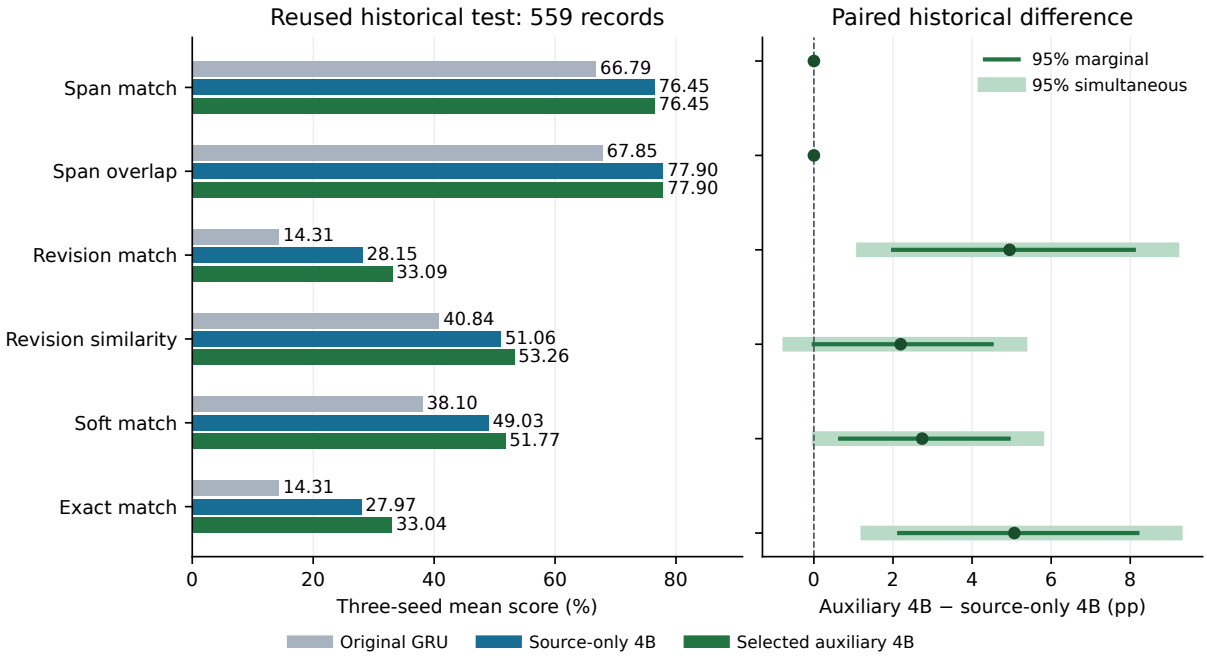


Fig. 3 Historical three-seed means (left) and matched auxiliary-minus-source differences (right). Intervals use 20,000 paired resamples of 194 available source units. Marginal and six-endpoint simultaneous intervals remain conditional on the fitted models and retained examples. Figure generated from archived numerical results.

5.3 Correction-pair and source-unit analysis

Figure 4 separates correction-pair familiarity from the full-set mean. On 499 pairs unseen in development, source-only and auxiliary 4B ExactMatch are 20.57% and 26.45%, a 5.88-point difference with a simultaneous interval of [1.70,10.52]. On the 60 seen-pair cases, the corresponding means are 89.44% and 87.78%. The seen-pair difference is negative, but its interval [-8.20,6.11] includes zero. The 5.07-point aggregate gain decomposes into approximately +5.25 points contributed by unseen pairs and -0.18 points contributed by seen pairs, weighted by subset sizes. This arithmetic describes where the gain occurs; it does not identify a causal mechanism or rule out other forms of overlap.

On the 207 cases without a shared available source unit, ExactMatch rises from 25.76% to 35.43%, a 9.66-point difference [2.10,18.41]. The wider interval reflects the smaller retained sample and clustering. This result reduces one documented overlap concern but does not establish learner-disjoint generalization: available identities are incomplete, the historical set has been reused, and this subset is part of the same evaluation archive.

5.4 Development controls for target fitting and fallback

Table 4 and Fig. 5 expose two decisions behind the selected pipeline. Auxiliary fitting without target adaptation is insufficient: its fallback reaches 18.00% original ExactMatch, below the source-only control's 25.50%. Adding source fitting after auxiliary exposure reaches 27.50%. Within that auxiliary-plus-source model, fallback raises original ExactMatch from the generator-only 25.00% to 27.50% and SoftMatch from 40.45% to 47.17%. On pooled stress views, ExactMatch increases from 21.25% to 24.875%. These controls support retaining both target fitting and the compact candidate in the selected implementation.

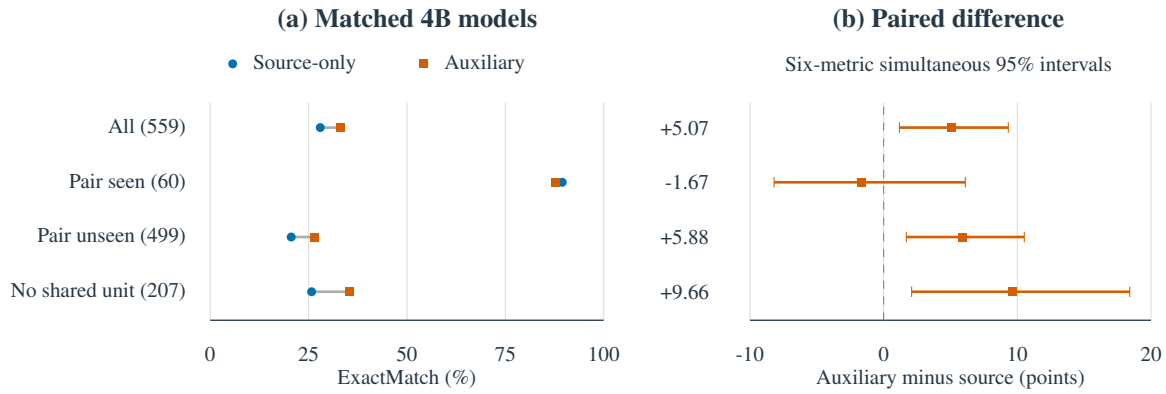


Fig. 4 Matched 4B ExactMatch by historical subset. Panel (a) shows three-seed means; panel (b) shows paired differences and six-endpoint simultaneous 95% intervals within each subset/comparator family. Sample sizes are examples. Available source-unit cluster counts are 194, 44, 190 and 95, respectively. Seen and unseen correction pairs partition the full set; the no-shared-unit subset overlaps these groups and is not an independent replication. Intervals do not adjust all adaptive studies or all subgroup comparisons jointly.

Table 4: Reused development controls (%). Orig.: 200 out-of-fold original contexts. Crop: 800 stress views from the same 200 parents, not 800 independent cases. Generator-only and fallback rows share saved locator offsets; legacy SM/SO score emitted correction records under the full scoring contract and can differ from standalone offset scores.

Variant	View	SM	SO	RM	RS	Soft	Exact
Source-only + fallback	Orig.	70.50	72.47	25.50	48.28	46.45	25.50
Source-only + fallback	Crop	69.38	71.52	22.13	45.73	43.16	21.88
Auxiliary only, generator	Orig.	47.00	48.43	9.00	25.64	24.61	9.00
Auxiliary only, generator	Crop	41.25	42.66	8.50	22.51	21.64	8.50
Auxiliary only + fallback	Orig.	70.50	72.47	18.00	42.04	40.85	18.00
Auxiliary only + fallback	Crop	69.38	71.52	18.75	41.90	40.45	18.75
Auxiliary + source, generator	Orig.	58.50	59.51	25.00	41.74	40.45	25.00
Auxiliary + source, generator	Crop	54.87	56.11	21.63	37.65	36.07	21.25
Auxiliary + source + fallback	Orig.	70.50	72.47	27.50	48.73	47.17	27.50
Auxiliary + source + fallback	Crop	69.38	71.52	25.25	46.59	44.53	24.88

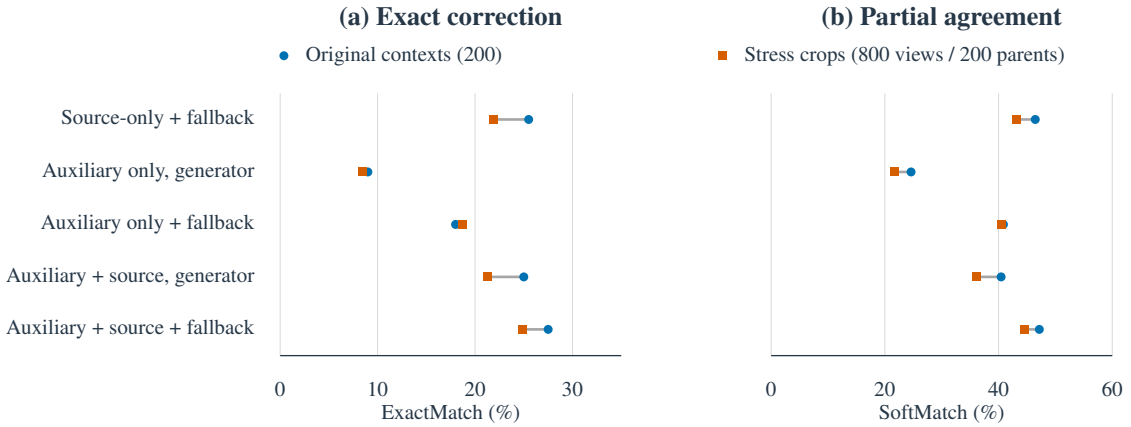


Fig. 5 Development controls under original and stress contexts. Points show means from the same fixed folds; connecting segments link each method's two conditions and do not denote learning trajectories. All five variants use the same saved localization outputs. The 800 crop views reuse 200 parent examples. These data participated in model selection and are descriptive, not independent confirmation.

The selected fallback still loses 2.625 points of ExactMatch and 2.638 points of SoftMatch between original and pooled stress contexts. Its gain over the source-only control is positive on both pooled conditions, but the individual left-positioned 64-character crop retains the 0.0766-point RS decline. Controlled crop robustness should therefore be distinguished from uniformly improved contexts and from transfer to naturally short learner text.

5.5 Unsuccessful controls and external transfer

The supplementary record retains all 21 development studies. A 695,689-parameter attention/copy repair model, surface marginalization, retrieval prompting, released-model task mappings, beam/edit selection, coverage extension, context gates and precision controls include unsuccessful outcomes. A released Japanese edit tagger's one-span mapping changes its original task and is not a reproduction of its published sentence-level evaluation.

Surface marginalization changes three development outputs and raises RS/SoftMatch by only 0.0690 points. Only 13 of 200 development outputs receive a numerical certificate, while 187 hit the search cap. On the historical set, ExactMatch is 16.99%, below fast fusion's 17.05%. Its small development advantage does not transfer. Exhaustive checks on 151 small lattices verify 2,166 asserted certificates across 4,077 decoder checks, establishing the recurrence on those cases rather than empirical correction superiority.

Teramura results reveal severe localization failure (Table 5). The context gate is evaluated after this external corpus had been inspected, so its result is a diagnostic rather than fresh confirmation. No correction score is inferred without literal replacement references.

Table 5: Teramura localization: 803 examples from 221 learners, three-seed means (%). Actual predicted offsets are scored.

Locator	Offset exact match	Offset overlap
Character GRU	1.04	7.59
Sparse linear	0.25	3.07
Fast fusion	0.12	2.08
Context-gated frontend	0.33	6.85

A descriptive audit finds median input lengths of 127.5 development characters versus 30 externally. Reference midpoints lie in the middle fifth for 92.00% of development targets but 13.20% externally. The sparse and fused predictions remain highly concentrated there. This association identifies distribution shift but does not isolate its cause.

All six earlier evaluated families obtain zero exact locations and corrections on the 25-case reserved style complement. The 4B model has not been evaluated on this complement. Its failure must therefore neither be assumed nor replaced by an unsupported success claim. The 803 localization records and 25 style records are distinct evaluations and are not pooled with the 559 historical corrections.

5.6 Resource costs and reproducibility

Fast fusion runs in a mean 13.6 ms per example on one CPU thread, including feature extraction and excluding loading. A portable implementation averages 11.3 ms and reproduces all 1,677 three-seed historical predictions; its archive is 14.7 MB. This is the measured CPU-interactive option.

The auxiliary 4B study uses one local 16 GB RTX 4070 Ti SUPER. Auxiliary fitting takes 2,504.18 seconds, with development-stage peak allocated/reserved memory reaching 9.788/15.777 GB. Its historical GPU pass takes 525.23 seconds and reaches 9.682/15.743 GB. These are study and batched-inference costs, not end-to-end singleton latency or minimum hardware requirements.

A separate, completed source-only 4B portable evaluation measures mean singleton latency of 0.538 seconds on 12 fixed development inputs, excluding loading. That value is not assigned to the auxiliary 4B pipeline. Full 4B CPU inference is unmeasured. Comparisons across 0.6B, 1.7B and 4B resource classes in the supplementary record retain their differing training exposure and frontends. The selected auxiliary model's portable delivery verification is not claimed complete here.

Table 6 keeps the measured resource tradeoff visible. The source-only 4B pipeline has about 2.33 times the stored values of the 1.7B pipeline and 1.81 times its measured GPU singleton latency. Its historical ExactMatch is only 1.19 points higher, while the matched auxiliary 4B model adds 5.07 points without changing stored size. These are descriptive ratios across nonidentical training histories; they neither isolate a capacity effect nor supply an unmeasured latency for the auxiliary pipeline. CPU measurements use float32 on one thread, whereas GPU measurements use a BF16 base with float32 adapters.

Table 6: Stored model values and measured singleton costs. Adapted-system timings use the same 12 development inputs and seed 20260927, include the frontend and exclude loading. Historical ExactMatch averages three seeds on a separate evaluation. A dash means unmeasured, not zero.

Pipeline	Values (10^9)	GPU (s)	CPU (s)	Exact (%)
0.6B, auxiliary 16k	0.605	0.270	10.141	21.65
0.6B, coverage 37,888	0.605	0.269	10.130	22.12
1.7B, auxiliary 16k	1.733	0.297	28.323	26.77
4B, source-only	4.043	0.538	—	27.97
4B, auxiliary 16k	4.043	—	—	33.04

6 Discussion and limitations

The matched 4B experiment supports auxiliary exposure as a useful training choice for this retrospective task. It improves exact annotated correction while retaining the same predicted location. Its uncertainty pattern is informative: exact correction gains are supported within the reported family, but incremental partial similarity remains uncertain. The evidence does not establish that additional data will help every subgroup or every task distribution.

The principal limitation is evaluation independence. Available source groups, correction-pair subsets, three seeds and paired intervals expose some dependencies but do not turn a reused source-overlapping benchmark into an untouched test. Extensive adaptive development adds uncertainty not captured by the reported intervals. A learner-disjoint full-correction evaluation is needed for stronger generalization claims.

The supplied-category, one-span contract is narrower than ordinary proofreading. One reference may penalize legitimate alternatives. Teacher-authored and automatic simulated errors differ from observed learner errors, and no audit establishes absence of source text from pretrained-model training. Natural context shift causes large localization failures, while the style complement shows failure even within an additional portion of a source corpus.

The strongest archived hosted backend reaches 40.61% ExactMatch, above the present 33.04%. Its complete original outputs are unavailable for independent rescoring, so it is contextual evidence rather than a new paired comparison. Scores of contemporary Japanese systems on other datasets are not interchangeable with these measures. Global state of the art, reliable classroom deployment, reduced teacher workload and learning benefits are not established.

7 Conclusion

Structured localization and sparse–recurrent fusion improve historical correction over a compact character GRU. At fixed 4B architecture, frontend and target-fitting conditions, auxiliary adaptation increases ExactMatch from 27.97% to 33.04%, with a positive six-metric-adjusted paired interval and unchanged localization. Partial-similarity gains against this matched control remain uncertain, and pair-seen examples retain regressions. The system offers a measured technical comparison under local compute, while external failures and historical-test reuse leave independent correction generalization unresolved.

Supplementary information

The main article provides five figures and six tables. The accompanying supplementary technical record preserves the full experiment history, component tables, unfavorable controls, data-construction audits, decoding details and resource comparisons from the baseline working manuscript. It provides detail beyond the focused main article; it is not an additional independent evaluation.

Statements and Declarations

Funding, competing-interest and author-identification statements are provided in the separate title-page file for double-blind review.

Data and code availability

The CAMC-JL release package contains 759 de-linked metadata records (200 development and 559 historical test cases), opaque grouping labels, development folds and 10,062 numerical evaluation records. Author-created metadata and results are licensed under CC BY 4.0; method code is MIT-licensed and author-trained frontend weights and three final adapters are Apache-2.0-licensed. The upstream Qwen base is obtained separately at a pinned revision. Learner sentences, gold spans and revisions, raw predictions and source-identity mappings are excluded; obtaining these materials remains subject to the source providers' conditions. The released numerical data support statistical reanalysis, but cannot independently reproduce text-level scoring or model fitting. The project homepage and repository have been prepared locally; public hosting and the final website URL remain pending verification. No completed public deposit is claimed in this draft. The supplementary technical record and source-access records remain available in the research archive.

Ethics

This work reanalyses existing learner-writing annotations and conducts no new recruitment, participant interaction or classroom intervention. It reports computational agreement with annotations rather than new human judgments or educational outcomes.

Use of generative AI

OpenAI Codex assisted with code development, experiment orchestration, analysis, manuscript drafting and revision, and document preparation. Reported numerical observations derive from executed experiments and archived outputs. The pretrained models evaluated in the study are described separately in Methods. Human authors remain responsible for verifying and approving the submitted manuscript.

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